Garrett Brownlee
PSY 4400 01
Dr. Sanford
10/01/25

Step #6 (Scale 1):

Calculate Cronbach's alpha

Console Terminal Background Jobs

R 4.5.1 · ~/Brownlee/

```
> listCERF <- c('emo_51.W', 'emo_54.W', 'emo_61.W', 'emo_64.W')
> alpha(x = proj2[ ,listCERF])

Reliability analysis
Call: alpha(x = proj2[, listCERF])

  raw_alpha std.alpha G6(smc) average_r S/N   ase mean   sd median_r
      0.82      0.82     0.8      0.53 4.6 0.019  2.6 0.94     0.59

    95% confidence boundaries
         lower alpha upper
Feldt     0.78  0.82  0.85
Duhachek  0.78  0.82  0.86

 Reliability if an item is dropped:
         raw_alpha std.alpha G6(smc) average_r S/N alpha se  var.r med.r
emo_51.W      0.82      0.82    0.75      0.60 4.5    0.020 0.0012  0.62
emo_54.W      0.80      0.80    0.74      0.58 4.1    0.022 0.0051  0.61
emo_61.W      0.74      0.74    0.70      0.49 2.9    0.028 0.0263  0.56
emo_64.W      0.72      0.73    0.68      0.47 2.7    0.031 0.0240  0.49

 Item statistics
           n raw.r std.r r.cor r.drop mean  sd
emo_51.W 257  0.76  0.75  0.63   0.55  2.5 1.2
emo_54.W 257  0.76  0.77  0.67   0.58  2.5 1.2
emo_61.W 257  0.84  0.85  0.78   0.71  2.7 1.1
emo_64.W 257  0.86  0.86  0.81   0.74  2.5 1.1

Non missing response frequency for each item
            0    1    2    3    4 miss
emo_51.W 0.09 0.15 0.21 0.32 0.24    0
emo_54.W 0.08 0.09 0.24 0.38 0.21    0
emo_61.W 0.05 0.11 0.15 0.45 0.23    0
emo_64.W 0.07 0.12 0.23 0.40 0.19    0
>
```

Garrett Brownlee
PSY 4400 01
Dr. Sanford
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Step #8 (Scale 1):

Calculate the mean and the standard deviation

Console Terminal Background Jobs

R 4.5.1 · ~/Brownlee/

```
> model2 <- lm(data=proj2, cerf_GB ~ 1)
> summary(model2)

Call:
lm(formula = cerf_GB ~ 1, data = proj2)

Residuals:
     Min       1Q   Median       3Q      Max 
-10.2179  -2.2179   0.7821   2.7821   5.7821 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  10.2179     0.2336   43.74   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.745 on 256 degrees of freedom

> |
```

Garrett Brownlee
PSY 4400 01
Dr. Sanford
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Step #9 (Scale 1):

A. Write: "mean = , standard deviation = , and standard error of measure = ," and fill in the blanks with the appropriate values for your target scale. You can calculate the standard error of measure based on your output.
   mean = 10.22
   standard deviation = 3.75
   standard error of measure = 1.59

B. Type a heading stating the name of your assigned scale, and then type the full text for all the questions in the scale you are analyzing.

   # Wife rating of her own hard emotion

   I felt angry.

   I felt annoyed.

   I felt irritated.

   I felt aggravated.

C. Following the prototype in the lecture 2 notes posted on Canvas, write a summary of the reliability and standard error of measure. When writing summaries, be sure to refer to the variables in your analysis using normal English words that people would understand even if they were not familiar with the dataset you are using. Never use the cryptic, abbreviated R variable names in your written summaries, and round all values to two decimal places.
   The wife rating of her own hard emotion scale had good reliability (Cronbach's Alpha = .82). These results suggest that 18% of the variance was measurement error and that on average, a participant's observed score is likely to deviate from her true score by 1.59 units.

Garrett Brownlee
PSY 4400 01
Dr. Sanford
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Step #10 (Scale 1):

"Scale 1" Scoring Table

1. **Step 6 screenshot. This screenshot should show the correct commands and results for a reliability analysis.**

Possible points	Deductions	Score
4	0	4

Criterion	Deduction
A. Section does not contain any content errors (missing steps or incorrect results).	2
B. Section contains no more than two content errors.	2
C. Screenshot size and resolution allow it to be read easily.	2
D. Screenshot cropped and does not show command lines with errors.	1
E. The correct set of variables has been combined to create a scale that matches an assigned scale.	4

2. **Step 8 screenshot. This screenshot should show the correct commands to create total score and use a linear model to calculate the mean and standard deviation for that total score.**

Possible points	Deductions	Score
4	0	4

Criterion	Deduction

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A. Section does not contain any content errors (missing steps or incorrect results).	2
B. Section contains no more than two content errors.	2
C. Screenshot size and resolution allow it to be read easily.	2
D. Screenshot cropped and does not show command lines with errors.	1
E. The student uses a linear model (and not some other method) to calculate the mean and standard deviation.	4

3. Step 9 parts A and B. This section includes information that needs to be placed prior to the written summary.

Possible points	Deductions	Score
6	0	6

Criterion	Deduction
A. The correct scale mean is listed.	2
B. The correct standard deviation is listed.	2
C. The correct standard error of measure is listed.	2
D. The full text for all the questions in the scale is provided.	2

4. Written summary. This section includes a written summary of the reliability and the standard error of measure.

Possible points	Deductions	Score
15	0	15

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PSY 4400 01
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Criterion	Deduction
A. The summary needs to name the scale being summarized. The name needs to be stated in plain language, and the summary should NOT contain any cryptic code names such as "FS_8."	**2**
B. The reliability needs to be correctly labeled as either poor, adequate, or good.	**4**
C. The correct Cronbach's Alpha value needs to be reported.	**4**
D. The summary needs to state the percent of variance that is measurement error.	**4**
E. The summary needs to make it clear that the student understands the meaning of a standard error of measure. This requires making a correct statement explaining that, on average, something deviates from something by some amount. This statement needs to make correct use of the words "observed" and "true" (or use other phrases that provide the same meaning and clarity), and when using these words, it needs to be clear whether the student is talking about something that belongs to a single person, a group of people, a sample, or a population.	**4**
F. The summary needs to use the correct value for the standard error of measure.	**2**
G. The summary needs to be written using concise sentences that are clear and easy to understand without extraneous or confusing information.	**2**
H. Values are rounded to two decimal places.	**2**

Garrett Brownlee
PSY 4400 01
Dr. Sanford
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5. Attachments.

Possible points	Deductions	Score
1.5	0	1.5

Criterion	Deduction
A. This grading table has been completed and is attached.	1.5

----------Steps 5-10 repeated below with Scale 2----------

Garrett Brownlee
PSY 4400 01
Dr. Sanford
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Step #6 (Scale 2):

Calculate Cronbach's alpha

Console Terminal Background Jobs

R 4.5.1 · ~/Brownlee/

```
> listCCI <- c('pbeh_71.H', 'pbeh_73.H', 'pbeh_75.H', 'pbeh_77.H', 'pbeh_82.H', 'pbeh_84.H', 'pbeh_86.H')
> alpha(x = proj2[ ,listCCI])

Reliability analysis
Call: alpha(x = proj2[, listCCI])

  raw_alpha std.alpha G6(smc) average_r S/N   ase mean
      0.85      0.85    0.85      0.45 5.8 0.014  2.2
   sd median_r
 0.88     0.45

    95% confidence boundaries
         lower alpha upper
Feldt     0.82  0.85  0.88
Duhachek  0.83  0.85  0.88

 Reliability if an item is dropped:
          raw_alpha std.alpha G6(smc) average_r S/N
pbeh_71.H      0.83      0.83    0.82      0.45 4.8
pbeh_73.H      0.83      0.83    0.81      0.45 4.9
pbeh_75.H      0.83      0.83    0.82      0.45 4.9
pbeh_77.H      0.82      0.82    0.81      0.43 4.6
pbeh_82.H      0.85      0.85    0.84      0.49 5.7
pbeh_84.H      0.84      0.84    0.83      0.46 5.2
pbeh_86.H      0.82      0.82    0.80      0.43 4.5
          alpha se  var.r med.r
pbeh_71.H    0.016 0.0077  0.44
pbeh_73.H    0.016 0.0066  0.45
pbeh_75.H    0.016 0.0103  0.48
pbeh_77.H    0.017 0.0103  0.42
pbeh_82.H    0.014 0.0060  0.50
pbeh_84.H    0.015 0.0100  0.45
pbeh_86.H    0.017 0.0076  0.43

 Item statistics
            n raw.r std.r r.cor r.drop mean   sd
pbeh_71.H 257  0.75  0.74  0.69   0.64  1.5 1.28
pbeh_73.H 257  0.75  0.73  0.69   0.63  2.0 1.31
pbeh_75.H 257  0.74  0.73  0.67   0.61  2.4 1.35
pbeh_77.H 257  0.77  0.78  0.73   0.67  2.4 1.20
pbeh_82.H 257  0.59  0.62  0.52   0.47  2.9 0.99
pbeh_84.H 257  0.68  0.69  0.62   0.56  1.9 1.10
pbeh_86.H 257  0.80  0.80  0.77   0.71  2.1 1.19

Non missing response frequency for each item
             0    1    2    3    4 miss
pbeh_71.H 0.24 0.34 0.16 0.16 0.10    0
pbeh_73.H 0.16 0.23 0.20 0.27 0.14    0
pbeh_75.H 0.11 0.19 0.16 0.27 0.27    0
pbeh_77.H 0.07 0.19 0.21 0.33 0.20    0
pbeh_82.H 0.02 0.09 0.14 0.48 0.27    0
pbeh_84.H 0.09 0.35 0.26 0.22 0.08    0
pbeh_86.H 0.07 0.29 0.23 0.26 0.15    0
```

Garrett Brownlee
PSY 4400 01
Dr. Sanford
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Step #8 (Scale 2):

Calculate the mean and the standard deviation

Console Terminal Background Jobs

R 4.5.1 · ~/Brownlee/

```
> model3 <- lm(data=proj2, cci_GB ~ 1)
> summary(model3)

Call:
lm(formula = cci_GB ~ 1, data = proj2)

Residuals:
     Min       1Q   Median       3Q      Max 
-15.2062  -4.2062  -0.2062   4.7938  12.7938 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  15.2062     0.3847   39.52   <2e-16 ***
---
Signif. codes:  
0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.168 on 256 degrees of freedom

>
```

Garrett Brownlee
PSY 4400 01
Dr. Sanford
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Step #9 (Scale 2):

A. Write: "mean = , standard deviation = , and standard error of measure = ," and fill in the blanks with the appropriate values for your target scale. You can calculate the standard error of measure based on your output.
mean = 15.21
standard deviation = 6.17
standard error of measure = 2.39

B. Type a heading stating the name of your assigned scale, and then type the full text for all the questions in the scale you are analyzing.

Husband rating of partner adversarial communication

My partner said something mean.

My partner raised his/her voice.

My partner told me that I was doing something to cause the problem.

My partner argued.

My partner defended his/her position.

My partner corrected my statements.

My partner criticized me.

C. Following the prototype in the lecture 2 notes posted on Canvas, write a summary of the reliability and standard error of measure. When writing summaries, be sure to refer to the variables in your analysis using normal English words that people would understand even if they were not familiar with the dataset you are using. Never use the cryptic, abbreviated R variable names in your written summaries, and round all values to two decimal places.
The husband rating of partner adversarial communication scale had good reliability (Cronbach's Alpha = .85). These results suggest that 15% of the variance was measurement error and that on average, a participant's observed score is likely to deviate from his true score by 2.39 units.

Garrett Brownlee
PSY 4400 01
Dr. Sanford
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"Scale 2" Scoring Table

6. Step 6 screenshot. This screenshot should show the correct commands and results for a reliability analysis.

Possible points	Deductions	Score
4	0	4

Criterion	Deduction
F. Section does not contain any content errors (missing steps or incorrect results).	2
G. Section contains no more than two content errors.	2
H. Screenshot size and resolution allow it to be read easily.	2
I. Screenshot cropped and does not show command lines with errors.	1
J. The correct set of variables has been combined to create a scale that matches an assigned scale.	4

7. Step 8 screenshot. This screenshot should show the correct commands to create total score and use a linear model to calculate the mean and standard deviation for that total score.

Possible points	Deductions	Score
4	0	4

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PSY 4400 01
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Criterion	Deduction
F. Section does not contain any content errors (missing steps or incorrect results).	2
G. Section contains no more than two content errors.	2
H. Screenshot size and resolution allow it to be read easily.	2
I. Screenshot cropped and does not show command lines with errors.	1
J. The student uses a linear model (and not some other method) to calculate the mean and standard deviation.	4

8. Step 9 parts A and B. This section includes information that needs to be placed prior to the written summary.

Possible points	Deductions	Score
6	0	6

Criterion	Deduction
E. The correct scale mean is listed.	2
F. The correct standard deviation is listed.	2
G. The correct standard error of measure is listed.	2
H. The full text for all the questions in the scale is provided.	2

Garrett Brownlee
PSY 4400 01
Dr. Sanford
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9. Written summary. This section includes a written summary of the reliability and the standard error of measure.	Possible points	Deductions	Score
	15	0	15

Criterion	Deduction
I. The summary needs to name the scale being summarized. The name needs to be stated in plain language, and the summary should NOT contain any cryptic code names such as "FS_8."	2
J. The reliability needs to be correctly labeled as either poor, adequate, or good.	4
K. The correct Cronbach's Alpha value needs to be reported.	4
L. The summary needs to state the percent of variance that is measurement error.	4
M. The summary needs to make it clear that the student understands the meaning of a standard error of measure. This requires making a correct statement explaining that, on average, something deviates from something by some amount. This statement needs to make correct use of the words "observed" and "true" (or use other phrases that provide the same meaning and clarity), and when using these words, it needs to be clear whether the student is talking about something that belongs to a single person, a group of people, a sample, or a population.	4
N. The summary needs to use the correct value for the standard error of measure.	2

Garrett Brownlee
PSY 4400 01
Dr. Sanford
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O. The summary needs to be written using concise sentences that are clear and easy to understand without extraneous or confusing information.	**2**
P. Values are rounded to two decimal places.	**2**

10. Attachments.

Possible points	Deductions	Score
1.5	**0**	**1.5**

Criterion	Deduction
B. This grading table has been completed and is attached.	**1.5**